

* understand cumulative TP, FP calculations & Precision Recall calculations

$$GT = 5$$

$$Predicted = 7$$

predictions already sorted by confidence.
so After 100 matching with G.T.

$$1 = \text{correct}$$

$$2 = \text{correct}$$

$$3 = \text{wrong}$$

$$4 = \text{correct}$$

$$5 = \text{wrong}$$

$$6 = \text{correct}$$

$$7 = \text{wrong.}$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

Cumulative TP FP

step 1 after Box 1

$$TP = 1 \quad FP = 0$$

$$\text{Precision} = 1 / 1 = 1$$

$$\begin{aligned} \text{Recall} &= 1 / 5 \\ &= 0.2 \end{aligned}$$

$$P = \frac{\text{correct}}{\text{total detected}}$$

total
correct detections.

$$R = \frac{\text{correct detections}}{\text{Actual (G.T.)}}$$

Step 2 After Box 2

$$TP = 2$$

$$FP = 0$$

$$P = 2/2 = 1$$

$$R = 2/5 = 0.4$$

$$P = \frac{\text{Correctly detected}}{\text{Total detected}}$$

Correctly detecting total

Step 3 After Box 3

$$TP = 2$$

$$FP = 1$$

$$P = 2/3 = 0.67$$

$$R = 2/5 = 0.4$$

$$R = \frac{\text{Correctly detected}}{G.T.}$$

Step 4 After Box 4

$$TP = 3$$

$$FP = 1$$

$$P = 3/4 = 0.75$$

$$R = 3/5 = 0.6$$

Step 5

$$TP = 3$$

$$FP = 2$$

$$P = 3/5 = 0.6$$

$$R = 3/5 = 0.6$$

step 6 After Box 6

$$TP = 4$$

$$FP = 2$$

$$P = 4/6 = 0.67$$

$$R = 4/5 = 0.8$$

$$P = \frac{\text{correct detected}}{\text{total detections}}$$

$$R = \frac{\text{correct detections}}{4.T}$$

step 7 After Box 7

$$TP = 4$$

$$FP = 3$$

$$P = 4/7 = 0.57$$

$$R = 4/5 = 0.8$$

step	TP	FP	P	R
1	1	0	1.00	0.20
2	2	0	1.00	0.40
3	2	1	0.67	0.40
4	3	1	0.75	0.60
5	3	2	0.60	0.60
6	4	2	0.67	0.80
7	4	3	0.57	0.80

observe

- * TP increases only when predictions matches a new ground truth.
- * FP increases when predictions is wrong or duplicate
- * Recall depends on GT count.
- * Precision drops when FP increases.